Integrated System Plan Feedback Loop Notice – HumeLink – 21 December 2023

AEMO publishes this notice pursuant to its function under clause 5.16A.5(b) of the National Electricity Rules (NER).

Context

AEMO's *Integrated System Plan* (ISP) is a roadmap for the energy transition in the National Electricity Market (NEM) over at least the next 20 years, in line with government policies to reach a net zero economy by 2050.

On 15 December, AEMO published the Draft 2024 ISP[1] and identified the optimal development path (ODP) – the lowest-cost pathway of essential generation, storage and transmission infrastructure to meet consumers' needs for secure, reliable and affordable electricity, and to achieve net zero emissions targets. In parallel, AEMO issued an update to the 2022 ISP[2] to update the ODP with that set out in the Draft 2024 ISP. The 2022 ISP as updated is currently the most recent ISP.

The ODP set out in the most recent ISP contains a series of power system investments, some of which are needed at or near their earliest delivery date which AEMO classifies as "actionable ISP projects". The ISP triggers the application of the Regulatory Investment Test for Transmission (RIT-T) for actionable ISP projects by RIT-T proponents (transmission network service providers (TNSPs) identified by AEMO)[3].

Following completion of the RIT-T, a TNSP may seek written confirmation from AEMO that the preferred option identified in the RIT-T remains aligned with the ODP in the most recent ISP and that the cost of the preferred option does not change the status of the actionable ISP project as part of the ODP. This process is referred to as the "feedback loop".

This AEMO confirmation via the feedback loop must be provided for a TNSP to be eligible to submit a contingent project application (CPA) to the Australian Energy Regulator (AER) for an actionable ISP project[4]. The AER's contingent project decision may adjust the TNSP's revenue allowance to reflect efficient and prudent forecast expenditure associated with the contingent project[5].

The HumeLink project

HumeLink is a proposed 500 kV line to reinforce the southern New South Wales network and connect the Snowy Mountains Hydroelectric Scheme and Project EnergyConnect to Bannaby. The project will provide access to increased generation and storage from Snowy Hydroelectric and renewable generation in Southern and Southwest New South Wales.

Regulatory approval processes for the HumeLink project completed under the ISP framework to date include[6]:

- 17 December 2021: Transgrid completed a RIT-T to assess the technical and economic viability of the project[7].
- 25 January 2022: Transgrid requested a feedback loop assessment ("feedback loop 1") that included early works costs of $383.3 million.
- 27 January 2022: AEMO confirmed the project satisfied the requirements of the feedback loop, including that the cost of early works did not change the status of the project as actionable.
- 5 April 2022: Transgrid submitted CPA (Stage 1, Part 1) with the AER for early works costs.

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[1] At https://aemo.com.au/consultations/current-and-closed-consultations/draft-2024-isp-consultation

[2] At https://aemo.com.au/en/energy-systems/major-publications/integrated-system-plan-isp/2022-integrated-system-plan-isp

[3] The RIT-T is a cost benefit analysis test that transmission network service providers (TNSPs) must apply to prescribed regulated investments in the transmission network. The purpose of the RIT-T is to identify the credible network or non-network options to address the identified network need that maximise net market benefits to the NEM. If an ISP identifies any actionable ISP projects, the ISP must specify the relevant RIT-T proponent which must apply the RIT-T in accordance with clause 5.22.6(a)(6) of the NER. RIT-T proponents (e.g. TNSPs) must apply the RIT-T to actionable ISP projects in accordance with rule 5.16A of the NER.

[4] A TNSP must obtain AEMO confirmation of the matters specified in clause 5.16A.5(b) of the NER. AEMO feedback loop assessments are available at: https://aemo.com.au/en/energy-systems/major-publications/integrated-system-plan-isp/integrated-system-plan-feedback-loop-notices.

[5] AER contingent project decisions are available at: https://www.aer.gov.au/industry/networks/contingent-projects.

[6] All costs provided in this feedback loop notice are expressed in real terms ($2022-23).

[7] Transgrid, *HumeLink RIT-T*, at https://www.transgrid.com.au/projects-innovation/humelink#RIT-T-process-and-submissions. On 17 December 2021, Transgrid released a Project Assessment Conclusions Report Addendum in response to a RIT-T dispute determination published by the Australian Energy Regulator.

- 17 August 2022: The AER approved $375.6 million in forecast capital expenditure (capex) to undertake early works.
- 6 April 2023: Transgrid requested a feedback loop assessment ("feedback loop 2") that expanded early works to include the procurement of long lead equipment.
- 19 May 2023: AEMO confirmed the project satisfied the requirements of the feedback loop, including that the total cost of early works ($632.9 million) did not change the status of the project as actionable[8].
- 23 May 2023: Transgrid submitted a CPA (Stage 1, Part 2) with the AER for the procurement of long lead equipment.
- 25 August 2023: The AER approved $227.9 million in forecast capex for the procurement of long lead equipment.

Feedback loop request

On 18 December 2023, Transgrid requested a third feedback loop assessment ("feedback loop 3") for the delivery of the HumeLink project containing the balance of project costs. The request included the following information relevant to this feedback loop assessment:

- The estimated total cost of the HumeLink project is $4.88 billion, which includes:
  - $4.28 billion for project delivery, and
  - $0.6 billion for early works activities already approved in CPAs by the AER.[9]
- The scope of the HumeLink project remains consistent with that considered and identified as preferred in the RIT-T.[10]
- Project timing (in service and full network capacity):
  - Northern Circuit (Gugaa to Bannaby): July 2026
  - Southern Circuit (Gugaa to Maragle to Bannaby): December 2026

Feedback loop assessment requirements

To be eligible to submit a CPA in relation to an actionable ISP project, a RIT-T proponent must obtain written confirmation from AEMO that:

- the preferred option addresses the relevant identified need specified in the most recent ISP and aligns with the ODP referred to in the most recent ISP; and
- the cost of the preferred option does not change the status of the actionable ISP project as part of the ODP (as amended by ISP update where applicable)[11].

Notice of AEMO confirmation that feedback loop requirements are satisfied

AEMO has undertaken this feedback loop assessment for the HumeLink project using the most recent ISP (i.e. the 2022 ISP as updated) and the cost estimates provided in Transgrid's feedback loop request. When considering the cost of the preferred option, AEMO assessed the total cost of the HumeLink project ($4.88 billion), including early works costs already approved by the AER in CPAs.

The ODP in the most recent ISP included the HumeLink project as an actionable ISP project. The project timing proposed in Transgrid's request for feedback loop 3 is consistent with that of an actionable project in service at its earliest delivery date. Transgrid's proposed project scope and total cost are also consistent with the scope and cost considered in the most recent ISP[12].

AEMO publishes this notice to confirm that:

- the HumeLink project addresses the relevant identified need and aligns with the ODP specified in the most recent ISP; and
- the total cost of the project, $4.88 billion ($2022-23), does not change the status of the actionable ISP project as part of the ODP specified in the most recent ISP.

[8] The total cost of early works assessed by AEMO in feedback loop 2 included early works costs approved by the AER in August 2022.

[9] Excludes equity raising costs of $33.14 million.

[10] Transgrid's PACR Addendum for the HumeLink project identified Option 3C as the RIT-T preferred option. Transgrid is currently updating its RIT-T analysis to confirm there is no change in the preferred option.

[11] Clause 5.16A.5(b) of the NER.

[12] Details are provided in Draft 2024 ISP, Appendix 5: Network Investments, p.22 and Appendix 6: Cost Benefit Analysis, p.40.